Practice Quiz: Perception (Computational Neuroscience)

Name: __ ***Date:*** __

Part A: Multiple Choice

1. In predictive coding, top-down signals from higher cortical areas carry: A) Raw sensory data B) Predictions about what lower areas should be processing C) Motor commands D) Emotional responses

2. Prediction errors in the cortex are primarily encoded in: A) Deep layers 5/6 B) Superficial layers 2/3 C) Layer 4 only D) The thalamus

3. The mismatch negativity (MMN) is: A) A brain response to any loud sound B) An automatic prediction error signal generated when an unexpected stimulus occurs in a repeating sequence C) Only found in humans D) A sign of brain damage

4. Gain control implements precision by: A) Making neurons physically larger B) Adjusting the sensitivity of neural populations to amplify or suppress prediction errors C) Adding new neurons D) Changing the speed of neural transmission

5. In the visual hierarchy, higher areas like IT cortex: A) Process basic features like edges and orientations B) Process complex features like objects and faces, sending predictions down to lower areas C) Only process color information D) Receive input directly from the retina

6. Repetition suppression (reduced neural activity with repeated stimuli) occurs because: A) Neurons get tired B) Predictions become more accurate, reducing prediction errors C) The brain stops processing the stimulus entirely D) Blood flow decreases

7. The canonical microcircuit is: A) Unique to the visual cortex B) A repeated wiring pattern found throughout the cortex implementing prediction and error computation C) A theoretical model with no neural evidence D) Only found in primates

Part B: Short Answer

1. Explain how the MMN experiment demonstrates that the brain is constantly predicting. What is being predicted, and how do we know the brain is predicting it even without conscious attention?

2. Draw or describe the flow of predictions and prediction errors through three levels of the visual hierarchy (e.g., V1, V2, V4) when you look at a familiar face. What information flows up? What flows down?

3. A patient reports vivid visual hallucinations despite having normal eye function. Using the predictive coding framework, propose a neural mechanism: Which component (predictions or prediction errors) might be malfunctioning, and how?